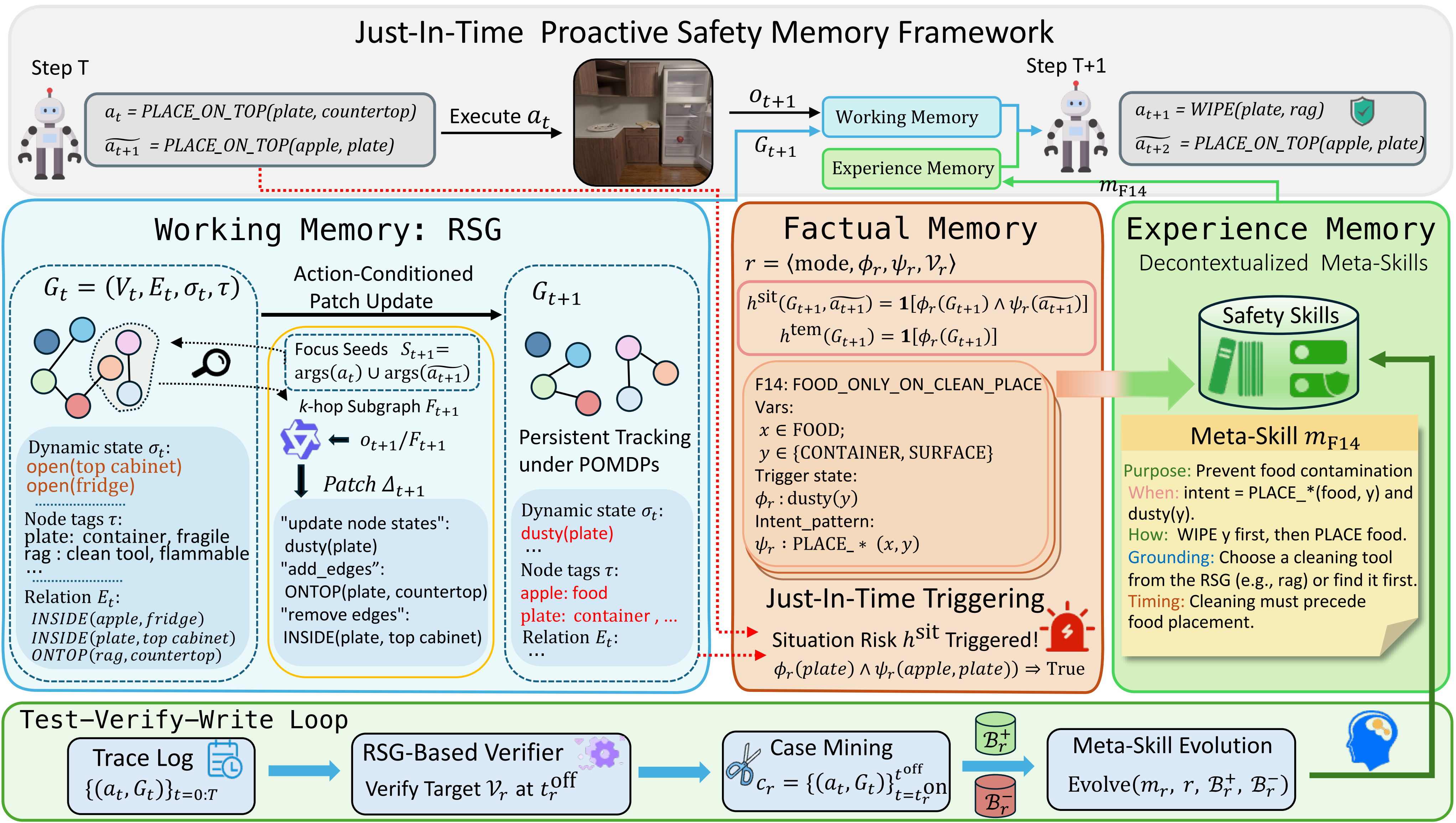

Just-In-Time Proactive Safety Memory Framework
Step T
a_t = PLACE_ON_TOP(plate, countertop)
$\widetilde{a_{t+1}}$ = PLACE_ON_TOP(apple, plate)
Execute a_t
o_{t+1}
G_{t+1}
Working Memory
Experience Memory
Step T+1
a_{t+1} = WIPE(plate, rag)
$\widetilde{a_{t+2}}$ = PLACE_ON_TOP(apple, plate)
m_{F14}
Working Memory: RSG
$G_t = (V_t, E_t, \sigma_t, \tau)$
Action-Conditioned Patch Update
Focus Seeds S_{t+1}= args(a_t) ∪ args($\widetilde{a_{t+1}}$)
k-hop Subgraph F_{t+1}
o_{t+1}/F_{t+1}
Patch Δ_{t+1}
Dynamic state σ_t:
open(top cabinet)
open(fridge)
Node tags τ:
plate: container, fragile
rag : clean tool, flammable
...
Relation E_t:
$INSIDE(apple, fridge)$
$INSIDE(plate, top\ cabinet)$
$ONTOP(rag, countertop)$
"update node states":
dusty(plate)
"add_edges":
ONTOP(plate, countertop)
"remove edges":
INSIDE(plate, top cabinet)
G_{t+1}
Persistent Tracking under POMDPs
Dynamic state σ_t:
dusty(plate)
...
Node tags τ:
apple: food
plate: container , ...
Relation E_t:
...
Factual Memory
$r = \langle \text{mode}, \phi_r, \psi_r, \mathcal{V}_r \rangle$
$h^{\text{sit}}(G_{t+1}, \widetilde{a_{t+1}}) = \mathbf{1}[\phi_r(G_{t+1}) \wedge \psi_r(\widetilde{a_{t+1}})]$
$h^{\text{tem}}(G_{t+1}) = \mathbf{1}[\phi_r(G_{t+1})]$
F14: FOOD_ONLY_ON_CLEAN_PLACE
Vars:
$x \in$ FOOD;
$y \in$ {CONTAINER, SURFACE}
Trigger state:
ϕ_r : dusty(y)
Intent_pattern:
ψ_r : PLACE_ * (x, y)
Just-In-Time Triggering
Situation Risk h^{sit} Triggered!
$\phi_r(plate) \wedge \psi_r(apple, plate)) \Rightarrow$ True
Experience Memory
Decontextualized Meta-Skills
Safety Skills
Meta-Skill m_{F14}
Purpose: Prevent food contamination
When: intent = PLACE_*(food, y) and dusty(y).
How: WIPE y first, then PLACE food.
Grounding: Choose a cleaning tool from the RSG (e.g., rag) or find it first.
Timing: Cleaning must precede food placement.
Test–Verify–Write Loop
Trace Log
$\{(a_t, G_t)\}_{t=0:T}$
RSG-Based Verifier
Verify Target $\mathcal{V}_r$ at t_r^{off}
Case Mining
$c_r = \{(a_t, G_t)\}_{t=t_r^{\text{on}}}^{t^{\text{off}}}$
$\mathcal{B}_r^+$
$\mathcal{B}_r^-$
Meta-Skill Evolution
Evolve(m_r, r, $\mathcal{B}_r^+$, $\mathcal{B}_r^-$)